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Date of preparation: 2026-04-15

Academic Appointments

2023-01 Associate Research Scientist
Present Columbia Aging Center

Mailman School of Public
Health, Columbia University

Education

2021-06 Ph.D., Biological Anthropology
*Tradeoffs between reproduction and aging
in the human epigenome*
Advisor: Christopher W. Kuzawa

Northwestern University

2018-06 M.A., Biological Anthropology

Northwestern University

2013-06 M.Sc., Biological Sciences

Simon Fraser University

2010-06 B.Sc.(Hons.), Zoology

University of Manitoba

Training

2022-02 Postdoctoral Research Scientist
2022-12

Mailman School of Public
Health, Columbia University

2021-02 Postdoctoral Researcher
2022-02

Department of Anthropology,
Northwestern University

Honors and Career Awards

2023-03	Best Poster International Workshop on Statistical Genetics	Boulder, Colorado, USA
2016-09	NSERC Postgraduate Scholarship <i>Accepted in place of the NSERC Alexander Graham Bell Canada Graduate Scholarship, reserved for top-ranked applicants, in order to carry out PhD in the USA</i>	Natural Sciences and Engineering Research Council of Canada (NSERC), Canada
2011-09	Michael Smith Foreign Study Supplement <i>Research conducted at The Roslin Institute, University of Edinburgh, UK</i>	Natural Sciences and Engineering Research Council of Canada (NSERC), Canada
2010-09	Alexander Graham Bell Canada Graduate Scholarship <i>Highest ranking graduate training grant in Canada</i>	Natural Sciences and Engineering Research Council of Canada (NSERC), Canada
2010-09	Provost Prize of Distinction	Dean of Graduate Studies, Simon Fraser University, Canada
2010-06	Harold E. Welch Award <i>Awarded for highest academic standing among all Honours students in the Bachelor of Science degree program</i>	Faculty of Science, University of Manitoba, Canada
2010-06	David Ian Mackenzie Medal <i>Awarded for highest academic standing among all students in the Zoology degree program</i>	Department of Zoology, University of Manitoba, Canada
2009-01	IDP Student Mobility Scholarship	IDP Education and University of Wollongong, Australia
2009-09	UMSU Scholarship for Excellence in Academic Achievement	University of Manitoba Students Union and University of Manitoba, Canada
2008	Undergraduate Summer Research Award (USRA)	Natural Sciences and Engineering Research Council of Canada (NSERC), Canada
2008	W.B. Scott Scholarship in Ichthyology	Huntsman Marine Science Centre, Canada
2007	Undergraduate Summer Research Award (USRA)	Natural Sciences and Engineering Research Council of Canada (NSERC), Canada

2007-06	Isbister Scholarship <i>Awarded to the 10 first-year university students with the highest academic standing</i>	Manitoba Scholarship and Bursary Initiative and University of Manitoba, Canada
2006-09	UMSU Scholarship for Excellence in Academic Achievement	University of Manitoba Students Union and University of Manitoba, Canada

Administrative Leadership and Academic Service

2023	Invited Scholar The Long-Term Effects of Pregnancy on Aging	National Institute on Aging Bethesda, Maryland, USA
2023	Invited Scholar Netherlands Institute for Advanced Study in the Humanities and Social Sciences (NIAS)	Early Health Disparities and Late Life Outcomes Amsterdam, Netherlands
2022	Meeting Reporter Canadian Institute for Advanced Research (CIFAR)	Child and Brain Development Vancouver, Canada
2021	Subcommittee on Aging Telomere Research Network	Tulane University NIA and NIEHS (NIH)
2019-	Meeting Co-organizer	Northwestern University
2020	Northwestern R User Group	Evanston, Illinois

Professional Organizations and Societies

Memberships and Positions

Member, American Association of Anthropological Genetics
 Member, American Association of Physical Anthropologists;
 Member, Human Biology Association;
 Member, International Society for Evolution, Medicine, and Public Health;
 Member, The Gerontological Society of America

Editorial Board

Associate Editor: *Epigenetics Reports*

Grant Reviewing

2025 National Medical Research Council (NMRC), Singapore
 2022 National Science Foundation (NSF), USA
 2016 The Israeli Science Foundation (ISF), Israel

Journal Reviewing

Advances in Geriatric Medicine and Research; Aging; Aging Cell; American Journal of Human Biology; Annals of Human Biology; Archives of Gynecology and Obstetrics; Behavioral Ecology; Behavioral Ecology and Sociobiology; Bioessays; BMC Bioinformatics; BMC Public Health; The BMJ; Cell Reports Medicine; Cell Metabolism; Clinical Epigenetics; Current Zoology; Demography; Diabetes Metabolism Research and Reviews; Environmental Research; Epigenetics; Evolution, Medicine, and Public Health; Evolutionary Applications; Exposome; Frontiers in Genetics; Frontiers in Zoology; GeroScience; Genome Medicine; Human Reproduction; Journal of Mammology; Journal of Molecular Evolution; Journals of Gerontology; Molecular Biology Reports; Mechanisms of Aging and Development; National Science Foundation; Nature Communications; Oncotarget; Proceedings of the National Academy of Sciences; Scientific Reports; Women's Health Report

Fellowship and Grant Support

Active Funding

2026-04	Alzheimer's Association Research Grant	US\$199,999
2029-07	Multi-Omic Links Between Reproductive Aging and AD Risk Among Midlife Women Role: Co-I (PI: Avila-Reiger) AARGA-26-1561061	
2026-02	Advanced Research Projects Agency	US\$10,554,642
2027-07	for Health (ARPA-H) Finding Aging Biomarkers by Searching Existing Trials (FAST) Role: Co-I (PI: Belsky) AY2AX000158	

- 2025-12 **National Institutes of Health (NIH)** US\$4,104,324
 2027-07 Long-term multi-omics follow-up of the CALERIE Trial
 to generate new knowledge for geroscience
 Role: Co-I (PI: Belsky)
 2R01AG061378
- 2025-04 **National Institutes of Health (NIH)** US\$4,134,258
 2030-03 Leveraging IVF to Identify Prenatal Effects Independent
 of Shared Maternal-Child Genes
 Role: Co-I (MPIs: Monk and Williams)
 1R01HD118949-01

Completed Funding

- 2024-12 **CUIMC Healthy Aging Initiative**
 2025-12 **(CHAI) Pilot Grant**
 Quantification of Biological Aging in Electronic Health
 Record Data for Precision Prognosis in ICU Patients
 Role: Co-I (PIs: Belsky and Moitra) US\$55,000
- 2021-06 **Telomere Research Network Pilot** US\$4,000
 2021-12 **Award (NIH/NIEHS)**
 Exploring the links between ovarian aging and somatic
 aging in humans: A study of leukocyte telomere length
 across the menopausal transition in a Philippine cohort
 Role: MPI w/ Kuzawa
- 2018-03 **National Science Foundation (NSF-DDIG)** US\$31,205
 2021-02 Quantifying the costs of human reproduction using the
 ‘Epigenetic Clock’
 Role: MPI w/ Kuzawa
 NSF1751912

Pending Funding

- Submit- **National Institutes of Health (NIH)/Department of** US\$4,022,330
 ted **Health and Human Services (DHHS)** Epigenetic sur-
 2025- rogates of reproductive history and ovarian senescence in
 10 Alzheimer’s disease risk
 Role: MPI w/ Avila-Reiger

Research Fellowships

2019	Data Science Fellowship Data Science for the Social Sciences Initiative Development Grant	Northwestern University
2018	Le Cron Foster and Friends of Anthropology Research Grant	Northwestern university
2018	Computational Essentials for Social Scientists Data Science for the Social Sciences Initiative	Northwestern University
2018	Intensive in Computational Evolutionary Medicine Evolutionary Medicine Summer Institute (EMSI)	Duke University
2015- 2016	Graduate Research Assistant Institute for Policy Research	Northwestern University
2012	Dean of Graduate Studies Fellowship	Simon Fraser University
2011	Dean of Graduate Studies Fellowship	Simon Fraser University

Educational Contributions

Direct Teaching

2022- 2025	Guest Lecturer, Life Course Epidemiology, Columbia University Mailman School of Public Health, Columbia University
2023- 2025	Invited Speaker, Children's Hospital Summer Internship Program, University of Colorado Anschutz Medical Campus
2024	Guest Lecturer, Environmental Epigenetics, Department of Anthropology, University of Texas Austin
2021	Guest Lecturer, Evolutionary Medicine, Department of Anthropology, Northwestern University
2020	Class Guest Author, Early Life Origins of Health and Wellbeing, Department of Anthropology, Emory University
2019	Workshop Assistant, Intro to R, R for Stata Users, Northwestern University, Research Computing Services
2019	Guest Lecturer, Evolutionary Medicine, Department of Anthropology, Northwestern University
2018	Teaching Assistant, Human Population Biology, Department of Anthropology, Northwestern University

- 2017 Guest Lecturer, Epigenetic Processes in Health and Evolution, Department of Anthropology, Northwestern University
- 2016 Guest Lecturer, Evolutionary Medicine, Department of Anthropology, Northwestern University
- 2012 Guest Lecturer, Endocrinology, Department of Biology, Simon Fraser University
- 2006-2009 Lab demonstrator and Tutor, Cell Biology, Department of Biology, University of Manitoba

Advising and Mentorship

- 2022-2025 Graduate Student Mentor (w/ Belsky), Columbia University Mailman School of Public Health, Columbia University

Co-mentored 20+ students in the Masters of Public Health and Masters of Biostatistics programs in data practices, problem solving, and professional development. Students have successfully transitioned to health research (e.g. New York Presbyterian Hospital, Massachusetts General Hospital), industry (e.g. CVS) and academia (e.g. PhD programs at Vanderbilt, UNC-Chapel Hill, and UT Austin).

- 2020-2022 Summer Undergraduate Research Assistant Fellowship (w/ Kuzawa), Institute for Policy Research, Northwestern University

Mentored two summer undergraduate students through the conceptualization, design, and analysis of projects related to epigenetic aging and early life development. Ms. Milan Hilde-Jones is now a graduate student at the Johns Hopkins Bloomberg School of Public Health. Mr. Raviraj Rege is studying medicine at Rice University. This work resulted in a manuscript published in *Clinical Epigenetics*.

- 2018-2020 Virtual Science Mentor, Mentor Matching Program, Illinois Science and Technology Institute (ISTI)

Provided guidance and feedback to high school students developing independent STEM research projects. Supported students in designing experiments, analyzing data, and communicating scientific findings. Encouraged problem-solving, critical thinking, and professional communication through virtual mentorship and project review sessions.

Educational Administration and Leadership

Program Development and Infrastructure

2022-2025 Training and Computational Infrastructure Development, Geroscience Computational Core

Designed and implemented a scalable onboarding and training system for biostatistics trainees. *Includes structured skills assessment, standardized coding practices, and reproducible workflow training*

Developed “Smart R: Practical Coding Habits for Data Science Success,” a training curriculum for data management, reproducibility, and collaborative workflows.

Established cloud-based collaborative data science environment (OneDrive + structured directory systems) enabling reproducible and shareable analysis pipelines.

Provide training in AWS-based computing environments (Ronin), including machine setup, data access, and large-scale omics analysis.

Founded and maintain the [Columbia Geroscience Computational Core GitHub](#), supporting shared code and cross-project collaboration.

Community Education or Training: Media Coverage and Interviews (Selected)

Biological Aging, Longevity, and Health

2024-06	5 Surprising Things That Make You Age Faster	AARP
2023-06	How to live longer: From cryonic freezing to DNA-repair research, longevity experts share their latest insights	Irish Independent
2023-02	The Science of Living Forever (or a Really, Really Long Time)	Medium.com

Reproduction and Biological Aging

2026-01	Parents Might Age Faster or Slower Based on How Many Kids They Have	Scientific American
2024-04	How Pregnancy May Speed Up the Aging Process	The Washington Post
2024-04	Just One Pregnancy Can Add Months to Your Biological Age	National Geographic
2024-04	Pregnancy Can Make You Age Faster	Time Magazine
2024-04	Having a Baby Can Age You on a Cellular Level	NBC Healthline
2018-10	Do Pregnancy and Childbirth Accelerate Aging in Women? Maybe.	The Washington Post

Caloric Restriction and Biological Aging

2023-02	Researchers May Have Found a Way to Slow Human Aging	Time Magazine
2023-02	Cut Calories and (Maybe) Add Years to Your Life	Bloomberg
2023-02	Restrict calories to live longer, study says, but critics say more proof is needed	CNN
2023-02	Cutting Calories Slows Aging	Medium.com

Early-life Environments and DNA Methylation

2019-04	Life in poverty greatly affects our DNA (translation)	BBC (Russia)
2019-04	Poverty Could Leave Its Mark on Genes	US News and World Report
2019-04	The link between socioeconomic status and genes	European Scientist
2017-09	Your Childhood Experiences Can Permanently Change Your DNA	Smithsonian Magazine

Patents and Inventions

Framingham-PACE	Ryan is an co-inventor of the FraminghamPACE epigenetic speedometer, patent pending. FraminghamPACE is a DNA methylation algorithm that measures the pace of biological aging. Working with Belsky and Geroscience Computational Core at Columbia, FraminghamPACE was developed in collaboration with the Moffitt-Caspi Lab at Duke University. The algorithm will be free to researchers. TruDiagnostic has secured the option for exclusive license for commercial applications.
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Publications

Peer-Reviewed Research Publications in Print or other Media

50. Warmbrunn MV, Yang L, Biswas RK, **Ryan CP**, Harper A, Belsky DW, Fiorito G, Fontana L. Weight regain reverses caloric restriction-induced benefits on the insulin-IGF-1 nutrient-sensing pathway: Post-hoc analysis from the CALERIE-2 randomized controlled trial. *Diabetes Care* (2026). dc251911.
49. Kim, K.I., Bello, N.A., Galea, L.A.M., Gladyshev, V.N., Jasienska, G., McKinley, K.L., Miller, E.C., Parks, W.T., Pollack, A.Z., Prewit, E.B., Rich-Edwards, J., **Ryan, C.P.**, Slaughter-Acey, J., Stanhewicz, A.E., Warrington, J.P., Suh, Y., 2025. A National Institute on Aging workshop on the long-term effects of pregnancy on aging. *Nature Aging* (2025) 1–4.
48. Balachandran, A., Pei, H., Shi, Y., Beard, J. R., Caspi, A., Cohen, A. A., Domingue, B. W., Eckstein Indik, C., Ferrucci, L., Furuya, A., Kothari, M., Moffitt, T. E., **Ryan, C. P.**, Skirbekk, V., Zhang, Y. S., & Belsky, D. W. Pace of Aging analysis of healthspan and lifespan in older adults in the US and UK. *Nature Aging* (2025), 5(6), 1132–1142.
47. Jude, M.S., Konwar, C., McQuaid, R.J., Ghani, F., Islam, N., Lalani, S., Merrill, S.M., Fatima, F., MacIsaac, J.L., Forchheh, N., **Ryan, C. P.**, Lee, N.R., Kuzawa, C.W., Kobor, M.S., Premji, S.S., Maternal-infant Global Health Team (MiGHT) - Collaborators in Research. Distinct maternal DNA methylation associations with gestational age at early and late-mid term pregnancy in a low- and middle-income country: evaluation of biological, genetic, and psychosocial contributors. *BMC Pregnancy and Childbirth* (2025), 25(1), 1062.
46. McDade, TW., Carrera, S., **Ryan, C. P.**, Burgner, D., Adair, L.S., Lee, N.R., Carba, D.B., MacIsaac, J.L., Dever, K., Kobor, M.S., Kuzawa, C.W. Evaluation of a DNA methylation-based measure of chronic inflammation in two generations of adults in metropolitan Cebu, Philippines. *Scientific Reports* (2025), 15(1): 32720.
45. Nyembwe, A., Zhao, Y., Caceres, B. A., Belsky, D. W., **Ryan, C. P.**, Taylor, B., Morrison, M. T., Prescott, L., Potts-Thompson, S., Aziz, A., Aruleba, F., Matute-Arcos, E., Williams, O., Crusto, C., & Taylor, J. Y. Discrimination, Coping, and DNAm Accelerated Aging Among African American Mothers of the InterGEN Study. *Epigenomes* (2025), 9(2), Article 2.
44. Zhuang, B. C., Jude, M. S., Konwar, C., Yusupov, N., **Ryan, C. P.**, Engelbrecht, H.-R., Whitehead, J., Halberstam, A. A., MacIsaac, J. L., Dever, K., Tran, T. K., Korinek, K., Zimmer, Z., Lee, N. R., McDade, T. W., Kuzawa, C. W., Huffman, K. M., Belsky, D. W., Binder, E. B., Czamara, D., Korthauer, K., Kobor, M. S. Accounting for differences between Infinium MethylationEPIC v2 and v1 in DNA methylation-based tools. *Life Science Alliance* (2025), 8(9).

43. **Ryan, C.P.**, Corcoran, D.L., Banskota, N., Eckstein Indik, C., Floratos, A., Friedman, R.A., Kobor, M.S., Kraus, V.B., Kraus, W., MacIsaac, J. and Orenduff, M.C., Pieper, C.F., White, J.P., Ferrucci, L., Horvath, S., Huffman, K.M., Belsky, D.W. The CALERIE Genomic Data Resource. *Nature Aging* (2024). 5, 320–331 <https://doi.org/10.1038/s43587-024-00775-0>.
42. Zhou J., Indik, C.E., Kuipers, T.B., Li, C., Nivard, M.G., **Ryan, C.P.**, Tucker-Drob, E.M., Taeubert, M.J., Wang, S., Wang, T., Conley, D., Heijmans, B.T., Lumey L.H., Belsky, D.W. Genetic analysis of selection bias in a natural experiment: Investigating in-utero famine effects on elevated body mass index in the Dutch Hunger Winter Families Study. *American Journal of Epidemiology* (2024). kwae376,
41. Savin, M.J., Wang, H., Pei, H., Aiello, A.E., Assuras, S., Caspi, T.E., Moffit, T.E., Muenning, C.P., **Ryan, C.P.**, Shi, B., Stern, Y., Sugden, K., Valeri, L., Belsky, D.W. Faster DunedinPACE, an epigenetic clock for pace of biological aging, is associated with accelerated cognitive aging among older adults in the Framingham Heart Study. *Alzheimer's & Dementia: Diagnosis, Assessment and Disease Monitoring* (2024). 16:e70031.
40. Cheng, M., Conley, D., Kuipers, T.B., Li, C., **Ryan, C.P.**, Taubert, M.J., Wang, S., Wang, T., Zhou, J., Schmitz, L.L., Tobi, E.W., Heijmans, B.T., Lumey, L.H., Belsky, D.W. Accelerated biological aging six decades after prenatal famine exposure. *Proceedings of the National Academy of Sciences of the United States of America* (2024). 121(24). e2319179121.
39. Domingo-Relloso, A., Riffo-Campos, A.L., Zhao, N., Ayala, G., Haack, K., Manterola, C., Rhoades, D.A., Umans, J.G., Fallin, M.D., Herreros-Martinez, M., Pollan, M., Boerwinkle, E., Platz, E.A., Jones, M.R., Bressler, J., Joehanes, R., **Ryan, C.P.**, Gonzalez, J.R., Levy, D., Belsky, D.W., and Tellez-Plaza, M. Multicohort Epigenome-Wide Association Study of All-Cause Cardiovascular Disease and Cancer Incidence: A Cardio-Oncology Approach. *JACC: CardioOncology* (2024). 6(5): 731-742.
38. **Ryan, C.P.**, Lee, N.R., Carba, D.B., MacIsaac, J.L., Lin, D.S., Atashzay, P., Belsky, D.W., Kobor, M.S., Kuzawa, C.W. Pregnancy is linked to faster epigenetic aging in young women. *Proceedings of the National Academy of Sciences of the United States of America* (2024). 121(16). e2317290121.
37. Hastings, W.J., Ye, Q., Wolf, S.E., **Ryan, C.P.**, Das, S.K., Huffman, K.M., Kobor, M.S., Kraus, W.E., MacIsaac, J., Martin, C.K., Racette, S.B., Redman, L.M., Belsky, D.W., Shalev, I. Effect of long-term caloric restriction on telomere length in healthy adults: CALERIE™ 2 trial analysis. *Aging Cell* (2024).
36. Bareja, A., Lee, D.E., Ho, T., Waitt, G., McKay, L.H., Hannou, S.A., Orenduff, M.C., McGreevy, K.M., Binder, A., **Ryan, C.P.**, Soderblom, E.J., Belsky, D.W., Kraus, V.B., Huebner, J.L., Kraus, W., Huffman, K.M., Baht, G.S., Horvath, S., Parmer, R.J., Miles, L.A., White, J.P. Liver-derived plasminogen mediates muscle

stem cell expansion during caloric restriction through the plasminogen receptor, Plg-RKT. *Cell Reports* (2024). 43(3). 113881.

35. Eisenberg, D.T.A., **Ryan C.P.**, Lee, N.R., Carba D.B., MacIsaac, J.L., Dever K., Atashzay P., Kobor, M.S., Kuzawa C.W. DNA methylation-based estimators of telomere length show low correspondence with paternal age at conception and other measures of external validity of telomere length. *GeroScience* (2024). 1-13.
34. Graf, G.H., Aiello, A.E., Kothari, M., Liu, H., Moffitt, T.E., Muennig, P., **Ryan, C.P.**, Sugden, K., Belsky, D.W. Educational Mobility, Pace of Aging, and Lifespan Among Participants in the Framingham Heart Study. *JAMA Network Open* (2024). 7(3):e240655.
33. Whitman, E.T., **Ryan, C.P.**, Abraham, W.C., Addae, A., Corcoran, D.L., Elliott, M.L., Hogan, S., Ireland, D., Keenan, R., Knodt, A.R., Melzer, T.R., Poulton, R., Ramrakha, S., Sugden, K., Williams, B.S., Zhou, J., Hariri, A.R., Belsky, D.W., Moffitt, T.E., Caspi, A. A blood biomarker of the pace of aging is associated with brain structure: replication across three cohorts. *Neurobiology of Aging* (2024). 136: 22-33.
32. Thomas, A., **Ryan, C.P.**, Caspi, A., Moffitt, T.E., Sugden, K., Zhou, J., Belsky, D.W., Gu, Y. Diet, pace of biological aging, and risk of dementia in the Framingham Heart Study. *Annals of Neurology* (2023). 95: 1069-1079.
31. McDade, T.W., **Ryan, C.P.**, Adair, L.S., Lee, N.R., Carba, D.B., MacIsaac, J.L., Dever, K., Atashzay, P., Kobor, M.S., and Kuzawa, C.W. Reply to Nwanaji-Enwerema, 2023, "Exposome and low birthweight selection bias considerations in early life infections and epigenetic age relationships". *American Journal of Human Biology* (2023). 35(11). e23977.
30. McDade, T.W., **Ryan, C.P.**, Adair, L.S., Lee, N.R., Carba, D.B., MacIsaac, J.L., Dever, K., Atashzay, P., Kobor, M.S., and Kuzawa, C.W. Association between infectious exposures in infancy and epigenetic age acceleration in young adulthood in metropolitan Cebu, Philippines. *American Journal of Human Biology* (2023). 35(11). e23948.
29. Waziry, R.[†], **Ryan, C.P.**[†], Corcoran, D.L., Huffman, K.M., Kobor, M.S., Kothari, M., Graf, G.H., Kraus, V.B., Kraus, W.E., Lin, D.T.S., Pieper, C.F., Ramaker, M.E., Bhapkar, M., Das, S.K., Ferrucci, L., Hastings, W.J., Kebbe, M., Parker, D.C., Racette, S.B., Shalev, I., Shilling, B., and Belsky, D.W. Effect of long-term caloric restriction on DNA methylation measures of biological aging in healthy adults from the CALERIE trial. *Nature Aging* (2023). (3):248-257.

[†]These authors contributed equally to the manuscript.

28. **Ryan, C.P.**, Rege, R.J., Lee, N.R., Carba, D.B., Kobor, M.S., MacIsaac, J.L., Lin, D.S., Atashzay, P., and Kuzawa, C.W. Maternal epigenetic clocks measured during pregnancy do not predict gestational age at delivery or offspring birth outcomes: a

- replication study in metropolitan Cebu, Philippines. *Clinical Epigenetics* (2022). 14(1):1-9.
27. **Ryan, C.P.**, Jones, M.J., Edgar, R.D., Lee, N.R., Kobor M.S., McDade, T.W., Kuzawa C.W. Immune cell type and DNA methylation vary with reproductive status in women: possible pathways for costs of reproduction. *Evolution, Medicine, and Public Health* (2022). 10(1):47-58.
 26. Kuzawa C.W., **Ryan, C.P.**, Adair, L.S., Lee, N.R., Carba, D.B., Kobor, M.S., MacIsaac, J.L., Dever, K., Atashzay, P., McDade, T.W. Birth weight and maternal energy status during pregnancy as predictors of epigenetic age acceleration in young adults from metropolitan Cebu, Philippines. *Epigenetics* (2022). 1-11.
 25. Johnson, A.P., **Ryan, C.P.**, Wallman, J.F. Climatic and biotic influences on the distributions of *Calliphora augur* and *Calliphora dubia* (Diptera: Calliphoridae). *Austral Entomology*. (2022). 61:370-377.
 24. **Ryan, C.P.** Epigenetic clocks: Theory and application in Human Biology. *American Journal of Human Biology*. (2020). e23488
 23. Shirazi, T.N., Hastings, W.J., Rosinger, A.Y. **Ryan, C.P.** Parity predicts biological age acceleration in post-menopausal, but not pre-menopausal, women. *Scientific Reports*. (2020). 10(1):20522
 22. **Ryan, C.P.** and Kuzawa, C.K. Germline epigenetic inheritance: Challenges and opportunities for linking human paternal experience with offspring biology and health. *Evolutionary Anthropology*. (2020). 1–21.
 21. Kuzawa, C.K., Bechayda, S.A., Borja, J.R.B., Carba, D.B., Duazo, P.L., Eisenberg, D.T.A., Georgiev, A.V., Gettler, L.T., Lee, N.R., Quinn, E.A., Rosenbaum, S. , Rutherford, J., **Ryan, C.P.**, McDade, T.W. Evolutionary life history theory as an organizing framework for cohort studies: Insights from the Cebu Longitudinal Health and Nutrition Survey. *Annals of Human Biology*. (2020). 47(2): 94-105.
 20. **Ryan, C.P.** Tandem Repeat Disorders. *Evolution, Medicine, and Public Health*. (2019). 1:17-17.
 19. McDade, T.W., **Ryan, C.P.**, Jones M.J., Hoke, M.K., Borja, J., Miller, G.E., Kuzawa C.W., Kobor M.S. Epigenetic signatures of socioeconomic status during development and early adulthood - *American Journal of Physical Anthropology*. (2019). 169(1):3-11.
 18. **Ryan, C.P.**, Hayes, M.G., Lee, N.R., McDade, T.W., Jones M.J., Kobor M.S., Kuzawa C.W., Eisenberg D.T.A. Reproduction predicts shorter telomeres and epigenetic age acceleration among young adult women. *Scientific Reports*. (2018). 8(1): 11100.

17. Georgiev, A.V., **Ryan, C.P.**, Borja, J.B., Kuzawa, C.W. Does growth during early postnatal “minipuberty” predict adult second-to-fourth digit ratio (2D:4D)? Evidence from Cebu, The Philippines. *American Journal of Human Biology*. (2017). 23(7): e23047.
16. McDade, T.W., **Ryan, C.P.**, Jones, M.J., MacIsaac, J.L., Morin, A.M., Meyer, J.M., Borja, J.B., Miller, G.E., Kobor, M.S., Kuzawa, C.W. Social and physical environments early in development predict DNA methylation of inflammatory genes in young adulthood. *Proceedings of the National Academy of Sciences of the United States of America*. (2017). 114: 7611-7616.
15. Georgiev, A.V., **Ryan, C.P.**, Gettler, L.T., McDade, T.W., Kuzawa, C.W. Second-to-fourth digit ratio (2D:4D) is unrelated to measures of somatic reproductive effort among young men from Cebu, The Philippines. *American Journal of Physical Anthropology*. (2017). 163(3):437–445.
14. **Ryan, C.P.**, Georgiev, A.V., Bechayda, S.A., McDade, T.W., Gettler, L.T., Eisenberg, D.T.A., Rzhetskaya, M., Hayes, M.G., Kuzawa, C.W. Androgen receptor polyglutamine repeat length (AR-CAGn) modulates the effect of testosterone on androgen- dependent somatic traits in Filipino young adult men. *American Journal of Physical Anthropology*. (2017). 163(2): 317–327.
13. Bairos-Novak, K.R., **Ryan, C.P.**, Freeman, A.R., Anderson, W.G., Hare, J.F. Like mother, like daughter: heritability of female Richardson’s ground squirrel (*Urocitellus richardsonii*) cortisol stress responses. *Current Zoology*. (2017). zox014. doi: 10.1093/cz/zox014
12. Gettler, L.T., **Ryan, C.P.**, Eisenberg, D.T.A., Rzhetskaya, M., Hayes, M.G., Feranil AB, Bechayda SA, Kuzawa, C.W. The role of testosterone in coordinating male life history strategies: The moderating effects of the androgen receptor CAG repeat polymorphism. *Hormones and Behavior*. (2017). 87: 164-175.
11. **Ryan, C.P.**, McDade, T.W., Gettler, L.T., Eisenberg, D.T.A., Rzhetskaya, M., Hayes, M.G., Kuzawa, C.W. Androgen receptor CAG repeat polymorphism and hypothalamic-pituitary-gonadal function in Filipino young adult males. *American Journal of Human Biology*. (2017). 29(1): e22897
10. **Ryan, C.P.**, Brownlie, J.C., and Whyard, S. Hsp90 and physiological stress are linked to autonomous transposon mobility in nematodes. *Genome Biology and Evolution*. (2016). 8 (12): 3794:3805.
8. **Ryan, C.P.**, Anderson, W.G., and Hare, J.F. Maternal gestational cortisol and testosterone are associated with trade offs in offspring sex and number in a free-living rodent (*Urocitellus richardsonii*). *PLoS One*. (2014). 9 (10): e111052.
7. Clary, D., Skyner, L.J., **Ryan, C.P.**, Gardiner, L.E., Anderson, W.G., and Hare, J.F. Shyness boldness, but not exploration, predicts glucocorticoid stress response

- in Richardson's ground squirrels (*Urocitellus richardsonii*). *Ethology*. (2014). 120: 1101-1109.
6. Hare, J.F., **Ryan, C.P.**, Enright, C., Gardiner, L.E., Skyner, L.J., and Anderson, W.G., Validation of a radioimmunoassay-based fecal corticosteroid assay for Richardson's ground squirrels (*Urocitellus richardsonii*) and behavioural correlates of stress. *Current Zoology*. (2014). 60 (5): 591–601
 5. **Ryan, C.P.**, Dawson, A.S., Sharp, P.J. and Williams, T.D. Circulating breeding and pre-breeding prolactin and LH are not associated clutch size in the Zebra Finch (*Taeniopygia guttata*) *General and Comparative Endocrinology*. (2014). 202: 26-34.
 4. **Ryan, C.P.**, and Crespi, B.J. Androgen receptor polyglutamine repeat tract length: models of selection and disease susceptibility. *Evolutionary Applications*. (2013). 6:180-196.
 3. Singh, A.D., Wong, S., **Ryan, C.P.**, Whyard, S. Oral delivery of double-stranded RNA in larvae of the yellow fever mosquito, *Aedes aegypti*; implications for pest mosquito control. *Journal of Insect Science*. (2013). 13.69.
 2. **Ryan, C.P.**, Anderson, W.G., Gardiner, L.E. and Hare, J.F. Stress-induced sex ratios in ground squirrels: support for a mechanistic hypothesis. *Behavioral Ecology*. (2012). 23:160-167
 1. Anderson, W.G., Dasiewicz, P.J., Liba, S., **Ryan, C.P.**, Taylor, J.R., Grosell, M., and Weihrauch, D. Gastro-intestinal handling of water and solutes in three species of elasmobranch fish, the white-spotted bamboo shark, *Chiloscyllium plagiosum*, little skate, *Leucoraja erinacea*, and the clear-nose skate *Raja eglanteria*. *Comparative Biochemistry and Physiology. Part A, Molecular and Integrative Physiology*. (2010). 155: 493-502.

Editorials

3. Belsky DW, **Ryan CP**. A daily multivitamin slows the ticking of epigenetic clocks. *Nature Medicine*. (2026). 32, 810–811 (2026).
2. **Ryan CP** and Belsky, DW. Epigenetic clock work ticks forward. *Nature Aging*. (2024). 4, 170-172.
1. **Ryan, C.P.** and Kuzawa, C.K. Aging: The temporary cost of dominance. *ELife*. (2021). 10. e68790.

Chapters

1. **Ryan, C.P.** Simple Sequence Repeats. In Vonk, J. and Shackelford, T.K. (eds.) *Encyclopedia of Animal Cognition and Behavior*. Springer International Publishing.

Invited and/or Peer-Selected Presentations at Local, Regional, National or International Levels

Invited Speaker: International

8. Ryan, C.P. (Speaker). Effect of long-term caloric restriction on DNA methylation measures of biological aging in healthy adults from the CALERIE trial. Foresight Institute: Biotech and Health Extension Group. Remote. January 9th, 2024.
7. Ryan, C.P. (Speaker). Quantifying Biological Aging Using Epigenetic Clocks. Office of Population Studies Foundation, University of San Carlos, Cebu City, Philippines. July 19th, 2022.
6. Ryan, C.P. (Speaker). Epigenetic Clocks in Primates and Other Species. Primate-omics Journal Club, Remote. August 4th, 2021.
5. Ryan, C.P. (Speaker). Tradeoffs Between Reproduction and Aging in the Human Epigenome. Département d'Anthropologie, Université de Montréal. Remote. March 19th, 2021.
4. Ryan, C.P. (Speaker). Using whole genome DNA methylation to study the costs of reproduction in a cohort of young Filipino women. Department of Biochemistry and Medical Genetics, University of Manitoba, Winnipeg, MB, Canada. January 24th, 2019.
3. Ryan, C.P. (Speaker). Evolution and Disease in the Human Androgen Receptor. Department of Biological Sciences Seminar Series, University of Manitoba, Winnipeg, MB, Canada. November 29th, 2013.
2. Ryan, C.P. (Speaker). Ryan, C.P. (Invited Speaker). Endocrine control of clutch-size determination in *Taeniopygia guttata*. Department of Behavioural Biology, Linnaeusborg Centre for Life Sciences, University of Groningen, the Netherlands. December 8th, 2011.
1. Ryan, C.P. (Speaker). Ryan, C.P. (Invited Speaker). Physiological orchestration of life history traits: Why do mechanisms matter? Center for Ecology and Hydrology, Edinburgh, Scotland, UK. November 29th, 2011.

Invited Speaker: National

10. Ryan, C.P. (Speaker). Cross-sectional and longitudinal changes in gravidity are associated with accelerated epigenetic age in young and middle-aged women in the Philippines. National Institute on Aging (NIA). Workshop on the Long-Term Effects of Pregnancy on Aging. Bethesda, MD, USA. November 1st, 2023.
9. Ryan, C.P. (Speaker). Cross-Sectional and Longitudinal Changes in Gravidity are Associated with Accelerated Epigenetic Age in Young and Middle-Aged Women in the Philippines. National Bureau of Economic Research (NBER), Cohort Studies Meeting. Los Angeles, CA, USA. January 19–21, 2023.

8. Ryan, C.P. (Speaker). Tradeoffs Between Reproduction and Aging in the Human Epigenome. Department of Anthropology, Binghamton University. Remote. March 3rd, 2022.
7. Ryan, C.P. (Speaker). Tradeoffs Between Reproduction and Aging in the Human Epigenome. Department of Anthropology, University of North Carolina Chapel Hill. Chapel Hill, North Carolina, USA. February 20th, 2022.
6. Ryan, C.P. (Speaker). Tradeoffs Between Reproduction and Aging in the Human Epigenome. Department of Anthropology, Pennsylvania State University. Remote. January 27th, 2022.
5. Ryan, C.P. (Speaker). Tradeoffs Between Reproduction and Aging in the Human Epigenome. Department of Anthropology, Emory University, Atlanta, GA, USA. February 3rd, 2020.
4. Ryan, C.P. (Speaker). Tradeoffs Between Reproduction and Aging in the Human Epigenome. Department of Anthropology, University of Minnesota Twin Cities, Minneapolis, MN, USA. January 27th, 2020.
3. Ryan, C.P. (Speaker). Tradeoffs Between Reproduction and Aging in the Human Epigenome. Department of Anthropology, Baylor University, Waco, TX, USA. November 20th, 2019.
2. Ryan, C.P. (Speaker). Evolution in Medicine, Human Reproduction, and Health. 22nd annual Britton C. and Lucille McCabe Lecture, Springfield College, Springfield, MA, USA. April 11th, 2019.
1. Ryan, C.P. (Speaker). Epigenetic Change As Biological Memory: Potential Pathways for Lifecourse Health Determinants in the Philippines. Department of Anthropology, Northwestern University, February 24th, 2017.

Oral Presentations at Professional Meetings

12. **Ryan, C.P.**, Lee, N.R., Carba, D.B., MacIsaac, J.L., Lin, D.S., Atashzay, P., Belsky, D.W., Kobor, M.S., McDade, T.W., Kuzawa, C.W. Cross-sectional and longitudinal changes in gravidity are associated with accelerated epigenetic age in young and middle-aged women in the Philippines. 48th Annual Meeting of the Human Biology Association. Reno, Nevada, U.S.A. April, 20th, 2023.
11. **Ryan, C.P.**, Rej, P.H., Holman, D.J., Lee, N.R., Hayes, M.G., Kuzawa, C.K., Eisenberg, D.T.A. Exploring the links between ovarian aging and somatic aging in humans: A study of leukocyte telomere length across the menopausal transition in a Philippine cohort. Telomere Research Network Annual Meeting, December 2nd, 2021. New Orleans, Louisiana, U.S.A.

10. **Ryan, C.P.**, Jones, M.J., Borja, J.B., McDade, T.W., Kuzawa, C.W. Testing for costs of reproduction using a new ‘epigenetic clock’ of all-cause mortality in a cohort of young women in the Philippines. Human Biology Association 45th Annual Meeting, April 15-17th, 2020. Los Angeles, California, U.S.A.
9. **Ryan, C.P.**, Kuzawa, C.W. Germline epigenetic inheritance: Challenges and opportunities of linking paternal experience with offspring biology and health. American Association of Physical Anthropologists and American Association of Anthropological Genetics, 44nd Annual Meeting, March 30th, 2019. Cleveland, Ohio, U.S.A
8. **Ryan, C.P.**, Edgar, R.D., Jones, M.J., Kobor, M.S., Borja, J.B., McDade, T.W., and Kuzawa C.W. Genome-wide DNA-methylation varies with reproductive status and supports costs of reproduction in a cohort of young Filipino women. International Society for Evolution, Medicine, and Public Health, 4th Annual Meeting, August 3rd. 2018. Park City, Utah, U.S.A.
7. **Ryan, C.P.**, Jones, M.J., MacIsaac, J.L., Morin, A., Borja, J.B., Kobor, M.S., Kuzawa, C.W., McDade, T.W. Epigenetic pathways of intergenerational phenotypic inertia in birth weight: Evidence from mothers in Cebu, Philippines. American Association of Physical Anthropologists and American Association of Anthropological Genetics, 42nd Annual Meeting, April 22nd, 2017. New Orleans, Louisiana, U.S.A.
6. **Ryan, C.P.**, Georgiev, A.V., McDade, T.W., Gettler L.T., Eisenberg, D.T.A., Hayes, M.G., Kuzawa, C.W. Variation in the androgen receptor modulates the relationship between testosterone and sexually dimorphic traits in adult males. Human Biology Association, 41st Annual Meeting, April 13-14, 2016. Atlanta, Georgia, U.S.A.
5. **Ryan, C.P.**, Dawson, A.S., Sharp, P.J. and Williams, T.D. Hormone-trait relationships for experimentally enlarged clutches continue to challenge the prolactin-based model for clutch-size determination. Society for Integrative and Comparative Biology, Annual meeting and exhibition, Jan. 3-7, 2013. San Francisco, California, U.S.A.
4. **Ryan, C.P.**, Dawson, A.S., Sharp, P.J. and Williams, T.D. Testing the prolactin-based mechanism for avian clutch-size determination in captive Zebra finches: Individual variability in hormone-trait relationships. North American Ornithological Conference, August 15-18, 2012. Vancouver, British-Columbia, Canada
3. **Ryan, C.P.**, Dawson, A.S., Sharp, P.J. and Williams, T.D. Experimental evidence for a role of prolactin in modulating avian clutch-size. Society for Integrative and Comparative Biology, Annual meeting and exhibition Jan. 3-7, 2012. Charleston, South Carolina, U.S.A.
2. **Ryan, C.P.**, Anderson, W.G., and Hare, J.F. Glucocorticoids, litter size optimization and sex allocation in *Urocitellus richardsonii*. Society for Integrative and

Comparative Biology, Annual meeting and exhibition, Jan. 3-8, 2011. Salt Lake City, Utah, U.S.A.

1. **Ryan, C.P.**, Anderson, W.G., and Hare, J.F. Struggling mothers, strong sons: optimization and adaptive sex allocation in Richardson's ground squirrels (*Spermophilus richardsonii*). Annual meeting for the Canadian Society of Zoologists, May 17-21, 2010. Vancouver, British Columbia, Canada

Poster Presentations at Professional Meetings (Selected)

6. **Ryan, C.P.**, Bose, P., Eckstein Indik, C., Belsky, D.W. The Columbia Aging Center Geroscience Computational Core (GCC). Healthspan Extension Summit. April 19th. 2024. New York, NY.
 5. **Ryan, C.P.**, Lee, N.R., Carba, D.B., MacIsaac, J.L., Lin, D.S., Atashzay, P., Belsky, D.W., Kobor, M.S., and Kuzawa, C.W. Pregnancy number is linked to faster epigenetic aging in young women, but not same-aged cohort men in the Cebu Longitudinal Health and Nutrition Survey (CLHNS). Healthspan Extension Summit. April 19th, 2024. New York, NY.
 4. [†]**Ryan, C.P.**, Lee, N.R., Carba, D.B., MacIsaac, J.L., Lin, D.S., Atashzay, P., Belsky, D.W., Kobor, M.S., and Kuzawa, C.W. Cross-sectional and longitudinal changes in gravidity are associated with accelerated epigenetic clock age in young and middle-aged women in the Philippines. International Workshop on Statistical Genetics, March 6-10th, 2023. Boulder, CO. U.S.A.
- [†]Awarded Best Poster at the International Workshop on Statistical Genetics, 2023.
3. **Ryan, C.P.**, Lee, N.R., Carba, D.B., MacIsaac, J.L., Lin, D.S., Atashzay, P., Belsky, D.W., Kobor, M.S., and Kuzawa, C.W. Cross sectional and longitudinal changes in gravidity are associated with accelerated epigenetic clock age in young and middle-aged women in the Philippines. New York City Epidemiology Forum, May 23rd, 2022. New York, NY, U.S.A.
 2. **Ryan, C.P.**, Lin, D.S., Kobor, M.S., McDade, T.W., Kuzawa, C.W. DNA methylation measurements using dried blood spots on the Illumina EPIC array yield high quality data comparable to whole blood measured using the 450k array. American Association of Physical Anthropologists, 45th Annual Meeting, April 15-17th, 2020. Los Angeles, California, U.S.A.
 1. **Ryan, C.P.**, Edgar, R.D., Jones, M.J., Kobor, M.S., Borja, J.B., McDade, T.W., Kuzawa, C.W. Genome-wide DNA-methylation in pregnancy and breastfeeding mark a 'landscape of adaptive immunity' and novel biological networks. Human Biology Association, April 12th, 2018. 43rd Annual Meeting, Austin, Texas, U.S.A.